PADHAKU
Exploration · Class 9 Science



CHAPTER 5

Exploring Mixtures and their Separation

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.
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🤔 Think It Over

1 **Why do suspended particles settle in muddy water over time but not in milk?**

✔ ANSWER

Mud particles are large (more than 1000 nm) and heavy, so gravity makes them settle — muddy water is a **suspension**.

Milk is a **colloid**: its particles are small (1–1000 nm) and stay evenly spread, so they do not settle.

2 **How is evaporation different from boiling?**

✔ ANSWER

- **Evaporation** happens at any temperature, only from the surface of the liquid, and slowly.
- **Boiling** happens at a fixed temperature (the boiling point), throughout the liquid, and fast (bubbles form).

3 **Why do you see bright rays of sunlight when it passes through small gaps between the leaves of a dense tree?**

✔ ANSWER

The air under the tree has tiny dust, smoke and water droplets. They **scatter** the sunlight, so the path of the light becomes visible. This is the **Tyndall effect**.

⭐ 5.1 How Can We Classify Mixtures?

? **Is the mixture of oil and water homogeneous or heterogeneous? Can you think of some other heterogeneous mixtures?**

✔ ANSWER

Oil and water is **heterogeneous** — they form two separate layers.

Other examples: sand and water, chalk powder in water, iron filings and sulfur, muddy water, smoke.

Act. 5.1 (4) **Activity 5.1: A = salt + water, B = chalk powder + water, C = few drops of milk + water. Are the particles visible in each mixture?**

✔ ANSWER

A: no particles visible (clear).

B: chalk particles are visible (cloudy, white).

C: particles not visible; the mixture looks slightly milky but uniform.

Act. 5.1 (5) **Pass a laser beam through each beaker and look from the side. What do you observe?**

✔ ANSWER

A: the path of the beam is **not visible**.

B: the path of the beam is **visible** (particles scatter light).

C: the path of the beam is **visible**.

Act. 5.1 (6) **Predict what you would observe in each beaker if you leave them undisturbed for a few minutes.**

✔ ANSWER

A: nothing settles.

B: chalk particles settle at the bottom.

C: nothing settles.

Act. 5.1 (7) **Filter each mixture separately. Is there any residue left on the filter paper?**

✔ ANSWER

A: no residue.

B: chalk powder is left as **residue**.

C: no residue (milk particles pass through filter paper).

Act. 5.1 (8) **Do you think these are the same types of mixtures or are they different?**

✔ ANSWER

They are **different**: A is a **solution**, B is a **suspension** and C is a **colloid**.

⭐ 5.2 Solutions

? **When farmers spray pesticides, if they do not mix the right amount of pesticide with water, what is likely to happen?**

✔ ANSWER

Too little pesticide may not protect the crops from pests.

Too much can damage the crops, soil and the environment, and waste money.

? **The right proportion is essential when preparing a solution. Can you think of other such examples?**

💡 **SAMPLE ANSWER**

- Medicines like cough syrup, saline drip (0.9% salt).
- Fertiliser solution for plants.
- Sugar syrup or lemonade, tea.
- Phenyl or disinfectant mixed with water for cleaning.

Fig. 5.4 **Collect some commercial packaged products and note down the information given for each item.**

💡 **SAMPLE ANSWER**

Example notes:

- Milk powder: protein, fat, sugar given in g per 100 g (% m/m).
- Glucose solution (drip): 5% m/v glucose.
- Vinegar: 5% v/v acetic acid.

⭐ Pause and Ponder · 5.2 Solutions

1 **A common talcum powder contains 4 % m/m zinc oxide. How much zinc oxide is present in 300 g of the talcum powder?**

✔ ANSWER

Given: % m/m = 4%, mass of powder = 300 g

Mass of zinc oxide = (4 ÷ 100) × 300 g

= 12 g

2 **Two tablespoons (15 mL each) of orange juice concentrate are mixed with water to make 150 mL of juice. What is the % v/v of concentrate?**

✔ ANSWER

Volume of concentrate = 2 × 15 mL = 30 mL

Volume of solution = 150 mL

% v/v = (30 ÷ 150) × 100

= 20 % v/v

3 **Vinegar contains 5 % v/v acetic acid. If you want to make vinegar from glacial (100%) acetic acid, how would you proceed?**

✔ ANSWER

5% v/v means 5 mL acetic acid in 100 mL of solution.

Take **5 mL** of glacial acetic acid (carefully, under supervision) and add water to make the total volume **100 mL**. Stir well.

For more vinegar keep the same ratio, e.g. 50 mL acetic acid made up to 1 L (1000 mL).

🧪 Activity 5.2 · Solubility curves

? **From Fig. 5.6, which compound, 'A' or 'B', will dissolve more in a given amount of water at a given temperature?**

✔ ANSWER

Compound **'B'** — its curve is always much higher than that of 'A'.

(i) **The solubility of compound 'A' in water at 20 °C is _____ (less than/more than/similar to) its solubility at 60 °C.**

✔ ANSWER

less than (slightly)

(ii) **The solubility of compound 'B' at 20 °C is _____ (less than/more than/similar to) its solubility at 60 °C.**

✔ ANSWER

less than

(iii) **The solubility of _____ increases more than that of _____ with an increase in the temperature.**

✔ ANSWER

The solubility of **compound 'B'** increases more than that of **compound 'A'**.

? **What do you think will happen if you make a saturated solution at a higher temperature and cool it slowly?**

✔ ANSWER

Solubility falls on cooling, so the extra solute cannot stay dissolved. It separates out as pure solid **crystals**.

⭐ 5.3.1 Crystallization

Act. 5.3 **Activity 5.3: Prepare a hot saturated solution of copper sulfate, filter it and let it cool slowly. What do you observe?**

✔ ANSWER

Large, shiny, well-shaped **blue crystals** of copper sulfate form as the solution cools.

Conclusion: a pure solid can be obtained from its saturated solution by **crystallization**.

You may also try **Place 1–2 mL of saturated copper sulfate solution on a glass plate and leave it. What do you observe? Did you get crystals? If yes, is this a good way to experiment? Explain.**

✔ ANSWER

Water evaporates and small blue crystals appear at the edges and on the plate.

Yes, crystals form, but they are **small, uneven and may be impure**, because the drop dries quickly and any dissolved impurities also stay behind on the plate.

So it is a quick way to see crystals, but not a good way to get large, pure crystals; slow cooling of a filtered saturated solution is better.

Think as a Scientist

Hypothesis: rapid cooling of hot saturated copper sulfate solution in ice-cold water gives smaller, less well-formed crystals than slow cooling at room temperature. How would you design an experiment to test this?

✔ ANSWER

- Prepare a hot saturated copper sulfate solution, filter it and divide it into **two equal parts** in identical beakers.
- Beaker 1: place in a tray of ice-cold water (rapid cooling).
- Beaker 2: cover and leave at room temperature (slow cooling).
- Keep everything else the same (volume, beaker, time).
- After some hours, compare crystal size and shape with a magnifying glass.
- If beaker 1 has smaller, irregular crystals, the hypothesis is supported.

⭐ Pause and Ponder · 5.3.1 Crystallization

4 **Using Fig. 5.6, if equal masses of hot saturated solutions of 'A' and 'B' are cooled from 80 °C to 60 °C, which solution is likely to deposit more solid?**

✔ ANSWER

Solution of **'B'**.

The solubility of 'B' falls a lot between 80 °C and 60 °C (steep curve), but that of 'A' hardly changes (nearly flat curve). So much more 'B' comes out as solid.

5 **Will there be any change in the size of common salt crystals if the rate of evaporation is increased or decreased? Explain.**

✔ ANSWER

Yes.

- **Slow evaporation** gives particles time to arrange in order → **larger**, well-shaped crystals.
- **Fast evaporation** → many **small**, less well-shaped crystals.

🧪 Activity 5.4 · Let us describe a process

Act. 5.4 **Fig. 5.9 shows how salt crystals are obtained from seawater (Seawater → Saturated solution → Salt crystals). Describe the process in your own words.**

✔ ANSWER

Seawater is let into shallow pits/pans near the sea. The Sun's heat and wind evaporate the water slowly, so the salt solution becomes more and more concentrated until it is **saturated**. As more water evaporates, salt separates out as **crystals**, which are collected, washed and dried.

⭐ 5.3.2 Distillation

? **When we evaporate a salt solution the solvent disappears. If we also want to recover the solvent, what should we do?**

✔ ANSWER

Use **distillation**: heat the solution, cool the vapour in a condenser and collect the pure liquid (water) in another vessel. Salt is left in the flask.

? **A student mixed acetone and water (two miscible liquids). How can we separate them? Can evaporation separate them and give both liquids?**

✔ ANSWER

No, evaporation would let the liquids escape into the air, so we would not get them back.

Use **distillation**: acetone (boils at about 56 °C) vaporises first, is cooled in the condenser and collected; water (100 °C) stays in the flask.

5.3.3 Paper Chromatography

? **What happens when a drop of water falls on something written with a sketch pen on paper?**

✔ ANSWER

The colour spreads. With a black sketch pen, different colours may appear as it spreads.

Act. 5.5 (5) **Observe the paper as the water rises through it. What do you notice?**

✔ ANSWER

The water carries the ink up the paper, and the ink spot spreads upwards.

Act. 5.5 (6) **As the water rises, the ink separates into different colour spots. What can you infer?**

✔ ANSWER

Black ink is a **mixture** of two or more coloured dyes. Each dye moves up at a different speed, so they get separated.

? **Will water work as a solvent in every case?**

✔ ANSWER

No. Water works only for components that dissolve in water. Pigments like those in spinach leaves do not dissolve in water, so we need alcohol or a mixture of solvents.

Pause and Ponder · 5.3 Methods of separation

6 (i) **Salt can be separated from a salt solution by evaporation or distillation.**

✔ ANSWER

True.

6 (ii) **Distillation can be used for separation of two liquids even when these have the same boiling point.**

✔ ANSWER

False. Correct: distillation separates liquids whose boiling points differ (by at least about 25 °C).

6 (iii) **In paper chromatography, the solvent level should be above the sample spot at the beginning of the experiment.**

✔ ANSWER

False. Correct: the solvent level should be **below** the sample spot, otherwise the spot dissolves into the solvent.

6 (iv) **Evaporation and crystallization are the same processes.**

✔ ANSWER

False. Correct: they are different. Evaporation removes all the solvent and leaves the solid (impurities too); crystallization forms pure crystals from a saturated solution on cooling.

⭐ 5.4.1 Separation of two immiscible liquids

Act. 5.6 (2) **Mustard oil and water are left to stand in a separating funnel. What do you observe?**

✔ ANSWER

They form **two separate layers**.

Act. 5.6 (3) **Mustard oil forms the upper layer and water the lower layer. Can you explain why?**

✔ ANSWER

Oil and water are **immiscible**, and mustard oil is **less dense** (lighter) than water, so it floats on top.

? **Can you think of any heterogeneous mixtures with a gas as one of the components?**

✔ ANSWER

Smoke (solid particles in air), fog (water droplets in air), dust in air, foam/froth (air bubbles in liquid).

What if ... **Two immiscible liquids of the same density are mixed in a separating funnel. How will the layers form?**

💡 SAMPLE ANSWER

Neither liquid is lighter, so they will not form a clear upper and lower layer quickly. Drops of one stay spread through the other (or the layers may form in no fixed order), so separating with a funnel would be hard.

⭐ 5.4.2 Sublimation

Act. 5.7 (6) **Heat a mixture of camphor and sand under an inverted funnel. Do you notice any solid deposits on the inner wall of the funnel?**

✔ ANSWER

Yes, **white solid camphor** deposits on the inner wall of the funnel; sand stays in the china dish.

Conclusion: camphor **sublimes** and can be separated from sand by sublimation.

? **Can you think of any other mixtures where sublimation can be used to separate the components?**

✔ ANSWER

- Naphthalene and common salt / sand
- Ammonium chloride and common salt
- Iodine and sand

? **Is it possible to dissolve one metal in another?**

✔ ANSWER

Not at room temperature. But when metals are **melted** at high temperature they mix to form a solution; on cooling it gives an **alloy** (e.g. brass, bronze).

⭐ Pause and Ponder · 5.4 Heterogeneous mixtures

7 **Why do immiscible liquids form two separate layers in a separating funnel?**

✔ ANSWER

They do not mix with each other and have **different densities**. The lighter liquid stays on top and the heavier one at the bottom.

8 **Is sublimation different from evaporation? Justify.**

✔ ANSWER

Yes.

- **Sublimation:** a **solid** changes directly into vapour without becoming liquid (e.g. camphor).
- **Evaporation:** a **liquid** changes into vapour from its surface, at any temperature.

⭐ 5.4.3 Suspensions

? **If muddy water is still not clear even after keeping for some time (and filtering), how can it be cleaned?**

✔ ANSWER

By **centrifugation** (fast spinning makes the fine particles settle) and/or **coagulation** (adding alum makes fine particles clump and settle).

Threads of Curiosity

The spinning game is called phugadi in Marathi and kikli in Punjabi. What is this called in your local language?

💡 **SAMPLE ANSWER**

Answer depends on your language, e.g. many Hindi-speaking areas just call it “ghoomna” or “chakri” game. Ask elders at home.

Bridging Science and Society

Find out more about simple technological inventions (like the paperfuge) that improve access to medical care in remote places.

💡 **SAMPLE ANSWER**

Examples: paperfuge, Foldscope (a paper microscope costing very little), solar-powered fridges for vaccines, rapid test kits for malaria.

Act. 5.8

Make your own centrifuge with a cardboard disc and thick thread. Which mixture would you like to separate using this mini centrifuge?

💡 **SAMPLE ANSWER**

For example muddy water or chalk powder in water in a small sealed tube. On spinning, the heavier particles move outwards and collect at the outer end.

? Can you think of any other coagulation processes used in everyday life?

✔ ANSWER

- Making paneer/cheese from milk with lemon juice or vinegar.
- Making curd from milk.
- Alum (fitkari) used to clear muddy water.
- Blood clotting at a cut.

⭐ 5.4.4 Colloids

? Blood can be separated by centrifugation and it coagulates — is it a suspension? But we cannot see blood cells with the naked eye — is it a solution?

✔ ANSWER

Neither. Blood is a **colloid**: its particles are not visible and do not settle on standing, but they can be separated by centrifugation.

? Can you think of some other substances that could be colloids?

✔ ANSWER

Fog, smoke, butter, cream, jelly, shaving foam, paint, cheese.

5.5 Tyndall Effect

? **Have you observed light scattering by particles in your surroundings? Think of some more examples.**

✔ ANSWER

- A beam of sunlight through a window or hole in a dark room.
- Car headlights in fog.
- Light from a cinema projector.
- Sunlight through gaps in leaves of a dense forest.

Pause and Ponder · 5.5 Tyndall effect

9 **Clouds are made up of tiny water droplets or ice crystals floating in the air. What type of mixture are clouds and why?**

✔ ANSWER

Clouds are a **colloid**. The droplets are tiny (colloidal size), they stay spread in the air without settling quickly, and clouds scatter light (Tyndall effect).

10 **Why do cities with a lot of smoke and dust in the air often look hazy?**

✔ ANSWER

Smoke and dust particles in the air **scatter light** (Tyndall effect). The scattered light makes the air look hazy and reduces how far we can see.

Activity 5.9 · Properties of mixtures

Table 5.1 **Complete Table 5.1: properties of solution, suspension and colloid.**

✔ ANSWER

Property	Solution	Suspension	Colloid
Nature	Homogeneous	Heterogeneous	Heterogeneous (looks homogeneous)
Particle size	Less than 1 nm	More than 1000 nm	1–1000 nm
Visibility	Not visible	Visible to naked eye	Not visible to naked eye
Separation by filtration	No	Yes	No
Settling	Does not settle	Settles on standing	Does not settle
Tyndall effect	No	Yes	Yes

Separation in daily life

? **Imagine trying to separate all the ingredients in a lemonade once they have been mixed. Can you do it?**

💡 SAMPLE ANSWER

Not easily. Water can be removed by distillation/evaporation, but sugar, salt and lemon juice components stay mixed together as solids and are very hard to separate from each other.

⭐ Revise, Reflect, Refine

1 **Which of the following mixtures are correctly classified as homogeneous (Hm) and heterogeneous (Ht)?**

✔ ANSWER

(iv) Muddy water — Ht, Milk — Ht, Blood — Ht, Brass — Hm

(i) is wrong (smoke is Ht), (ii) is wrong (brass, vinegar are Hm; muddy water is Ht), (iii) is wrong (milk is Ht).

2 **Which mixtures show the Tyndall effect? Explain the reason for correct and incorrect options.**
(a) air and dust particles (b) copper sulfate and water (c) starch and water (d) acetone and water
(i) a and b (ii) b and d (iii) a and c (iv) c and d

✔ ANSWER

(iii) a and c

- (a) Dust in air and (c) starch in water have large (colloid-size) particles that scatter light → Tyndall effect.
- (b) Copper sulfate in water and (d) acetone in water are true solutions; their particles are too small (< 1 nm) to scatter light.

So options (i), (ii) and (iv) are wrong because each contains a solution.

3 **Use the words/phrases to fill Table 5.2 (properties and examples of solution, suspension, colloid).**

Large-sized particles; Particles remain evenly distributed; Small-sized particles (less than 1 nm); Moderate-sized particles (1–1000 nm); Settles down when left undisturbed (more than 1000 nm); Does not settle down; Scatters light; Separates by filtration; Transparent; Salt solution; Milk; Sand in water; Smoke; Heterogeneous mixture; Cannot be separated by filtration; Mud; Butter; Brass.

✔ ANSWER

Solution	Suspension	Colloid
Small-sized particles (less than 1 nm)	Large-sized particles	Moderate-sized particles (1–1000 nm)
Particles remain evenly distributed	Settles down when left undisturbed (more than 1000 nm)	Particles remain evenly distributed
Does not settle down	Heterogeneous mixture	Does not settle down
Transparent	Scatters light	Scatters light
Cannot be separated by filtration	Separates by filtration	Cannot be separated by filtration; Heterogeneous mixture
Examples: Salt solution, Brass	Examples: Sand in water, Mud	Examples: Milk, Smoke, Butter

4 (i) A cake recipe uses 75 g sugar, 420 g all-purpose flour and 5 g sodium hydrogencarbonate. Express the concentration of each component using an appropriate method.

✔ ANSWER

All are solids measured by mass, so use **% m/m**.

Total mass = 75 + 420 + 5 = 500 g

Sugar = (75 ÷ 500) × 100 = **15 % m/m**

Flour = (420 ÷ 500) × 100 = **84 % m/m**

Sodium hydrogencarbonate = (5 ÷ 500) × 100 = **1 % m/m**

4 (ii) **A brass alloy contains 70% copper by mass. Calculate the quantities of copper and zinc in 120 g of brass.**

✔ ANSWER

Copper = (70 ÷ 100) × 120 g = **84 g**

Zinc = 120 g − 84 g = **36 g** (30%)

5 **A cooking oil pack says one litre (910 g). If this oil is mixed with water, will it form a separate layer? Which will be on top? How will you separate the layers? Draw the apparatus.**

✔ ANSWER

Density of oil = 910 g ÷ 1000 mL = 0.91 g/mL; water = 1 g/mL.

Yes, oil and water are immiscible, so they form **two layers**. Oil is lighter, so **oil is on top**.

Separate them with a **separating funnel**: let it stand, open the stopcock to drain the water (lower layer), discard the small mixed portion, then collect the oil.

Diagram: separating funnel on a laboratory stand with stopcock, oil layer above water layer, conical flask below (as in Fig. 5.16).

6 **Assertion (A): Solutions do not exhibit the Tyndall effect.**

Reason (R): The particles in solutions are larger than 100 nm, so they cannot scatter light.

✔ ANSWER

(iii) A is true, but R is false.

Solution particles are very small (less than 1 nm); that is why they cannot scatter light.

7 **How would you separate the mixtures in Table 5.3? Give the reason for your method.**

✔ ANSWER

Mixture	Method	Reason
Mud from muddy water	Sedimentation and decantation/filtration; coagulation with alum or centrifugation for fine particles	Mud is insoluble and heavier than water
Plasma from blood	Centrifugation	Heavier blood cells move to the bottom on fast spinning; plasma stays on top
Naphthalene and sand	Sublimation	Naphthalene sublimes, sand does not
Chalk powder and common salt	Dissolve in water, filter, then evaporate the filtrate	Salt dissolves in water, chalk does not
Common salt and water	Evaporation (or distillation to get water also)	Water vaporises, salt is left behind

Oil from water	Separating funnel	Immiscible liquids of different densities
Pigments of the flower	Paper chromatography	Pigments move at different speeds on the paper

8 **Two miscible liquids A (b.p. 60 °C) and B (b.p. 90 °C) are mixed. Suggest a method to separate them and draw a labelled diagram.**

✔ ANSWER

Difference in boiling points = 90 − 60 = 30 °C (more than 25 °C), so use **distillation**.

Heat the mixture; A boils first, its vapour is cooled in the condenser and collected; B remains in the flask.

Diagram labels (as Fig. 5.12): distillation flask with mixture, thermometer, water condenser (water inlet/outlet), conical flask with distillate, wire gauze, tripod stand, burner, laboratory stand.

9 **Compare evaporation, crystallization and distillation. In which situation would you prefer each?**

✔ ANSWER

- **Evaporation:** solvent is lost to air; solid left behind (impurities too). Use when only the dissolved solid is needed and purity is not important, e.g. salt from seawater.

- **Crystallization:** pure crystals form from a saturated solution on cooling; impurities stay in solution. Use to get a **pure** solid, e.g. copper sulfate, sugar.
- **Distillation:** liquid vaporised and condensed back. Use when the **liquid is also needed** or to separate miscible liquids with boiling points differing by about 25 °C or more, e.g. acetone and water, pure water from salt water.

10 **Blood is a colloidal mixture. (i) What would happen if blood behaved like a true suspension inside the body? (ii) Identify the dispersed phase and the dispersion medium in blood.**

✔ ANSWER

(i) Blood cells would **settle down** in the vessels; blood would not stay evenly mixed, so it could not carry oxygen and nutrients properly and might block blood vessels.

(ii) **Dispersed phase:** blood cells (red, white blood cells) and platelets. **Dispersion medium:** plasma (mostly water).

11 **A mixture of sand, common salt and naphthalene is separated using the steps in Fig. 5.25b (1: heating under an inverted funnel, 2: heating a solution in a china dish, 3: filtration). Write the correct sequence.**

✔ ANSWER

1 → 3 → 2

- Step 1: **Sublimation** — naphthalene collects on the funnel.

- Add water to the rest; salt dissolves.
- Step 3: **Filtration** — sand stays on the filter paper.
- Step 2: **Evaporation** of the filtrate — salt is left in the china dish.

12 **Why is distillation an effective method for separating a mixture of water and acetone?**

✔ ANSWER

Acetone boils at about 56 °C and water at 100 °C — a large difference (44 °C, more than 25 °C). So acetone vaporises first and is condensed and collected, while water stays in the flask.

13 **Table 5.4: Solubility (g per 100 g water)**

Salt	10 °C	20 °C	30 °C	40 °C	60 °C	80 °C
Potassium nitrate	21	32	45	62	106	167
Sodium chloride	36	36	36.3	36.5	37	37
Potassium chloride	35	35	37.4	40	46	54
Ammonium chloride	24	37	41	41	55	66

✔ ANSWER

See parts (i)–(iii).

13 (i) **What mass of potassium nitrate would be needed to prepare its saturated solution in 50 g of water at 40 °C?**

✔ ANSWER

At 40 °C: 62 g in 100 g water.

In 50 g water = 62 × 50 ÷ 100

= 31 g

13 (ii) **A saturated solution of potassium chloride at 80 °C is left to cool to room temperature (25 °C). What would she observe? Explain.**

✔ ANSWER

Crystals of potassium chloride appear as it cools. Solubility falls from 54 g (80 °C) to about 36 g (25 °C) per 100 g water, so about 18 g per 100 g water can no longer stay dissolved and separates out.

13 (iii) **What is the effect of temperature on solubility of salts? Compare the four salts from 10 °C to 80 °C.**

✔ ANSWER

Solubility of these salts **increases** with temperature, but by different amounts:

- Potassium nitrate: 21 → 167 g (largest increase)
- Ammonium chloride: 24 → 66 g
- Potassium chloride: 35 → 54 g
- Sodium chloride: 36 → 37 g (almost no change)

14 **Student A: 20 g sugar in 80 g water; B: 20 g sugar in 100 g water; C: 30 g sugar in 80 g water.**
(i) Calculate % m/m of each. (ii) Whose solution is most concentrated? Why?

✔ ANSWER

(i) % m/m = mass of solute ÷ mass of solution × 100

A: 20 ÷ (20 + 80) × 100 = **20 %**

B: 20 ÷ (20 + 100) × 100 = **16.7 %**

C: 30 ÷ (30 + 80) × 100 = **27.3 %**

(ii) **Student C** — it has the most sugar per 100 g of solution (highest % m/m).

15 **Examine Fig. 5.26 (heated flask A, long tube with water B, collecting flask C).**
(i) Identify technique 'S'. (ii) Label A, B and C. (iii) Which mixtures can be separated by it?
Boiling points: water 100 °C, acetone 56 °C, alcohol 78 °C, chloroform 61 °C, benzene 80 °C.
(a) water–acetone (b) water–salt (c) acetone–alcohol (d) sand–salt (e) alcohol–chloroform (f) alcohol–benzene

✔ ANSWER

(i) S = **Distillation**.

(ii) A = **distillation flask**, B = **(water) condenser**, C = **conical flask** (receiver, collects distillate).

(iii) **(a) water–acetone** (difference 44 °C) and **(b) water–salt** (water distils, salt stays).

Not (c) 22 °C, (e) 17 °C, (f) 2 °C — differences less than 25 °C (need fractional distillation); not (d) — sand and salt are both solids.

🎉 **Chapter 5 done! Revise the answers once more tomorrow.**